

TUSHAR PANTHRI

(934) 263-3282 | tusharpanthri@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | New York, NY(open to relocation)

PROFESSIONAL SUMMARY

Ownership-driven Software Engineer building scalable, fault-tolerant distributed systems with cloud-native microservices on AWS. Demonstrated end-to-end ownership across design, CI/CD deployment, CloudWatch monitoring, and on-call incident resolution for high-availability production services. Collaborative Agile practitioner leveraging GenAI development tools and modern SDLC best practices to ship clean, maintainable Python and Java code with measurable customer impact.

EXPERIENCE

American Express

October 2023 – August 2024

Software Engineer

Gurugram, Haryana

- Designed and developed fault-tolerant backend services using Python and SparkSQL for 18 U.S. channels, applying distributed systems patterns and RESTful microservices to ensure scalable, cost-effective data processing at scale
- Owned the end-to-end lifecycle of ETL microservices on cloud infrastructure, implementing CI/CD via GitHub Actions, automated testing, and CloudWatch monitoring to improve release consistency and incident resolution
- Collaborated cross-functionally in Agile squads to integrate GenAI-aware data signals into customer-facing products, writing clean OOP-compliant Python code and contributing technical documentation

Ameriprise Financial

May 2022 – October 2023

Software Engineer

Gurugram, Haryana

- Architected cloud-native data pipelines on AWS using Spark and Lambda, optimizing parallel processing and microservices communication to reduce pipeline latency by ~80% while maintaining resilient, scalable system design
- Built fault-tolerant backend services with automated observability via CloudWatch and SNS alerts, applying TDD and SDLC best practices to ensure high availability for the BlackRock Aladdin analytics platform under production load
- Mentored junior engineers on distributed systems design patterns, code review standards, and AWS cloud best practices, fostering an ownership mindset and rapid adaptation to new technologies in a fast-paced engineering culture

Cognizant Technology Solutions

June 2020 – May 2022

Software Developer

Pune, Maharashtra

- Built Python and Java backend modules for enterprise insurance and financial domain client applications following Agile/Scrum delivery cycles.
- Wrote unit and integration tests to validate features and catch regressions; resolved production defects via log-based root-cause analysis in collaboration with QA and business analysts.

PROJECTS

Scalable Distributed Transaction Processing System | *Go, Multi-Paxos, 2PC, gRPC*

- Sharded, fault-tolerant transaction engine in Go using Multi-Paxos leader election and Two-Phase Commit over gRPC.
- Implemented configurable sharding and tunable consistency with crash recovery on node failure, verifying linearizable state-machine replication while sustaining 1,000+ transactions/sec across three clusters.

Byzantine Fault-Tolerant Banking System | *Go, PBFT, gRPC*

- Byzantine fault-tolerant banking backend in Go using linear-PBFT over gRPC, tolerating f faulty replicas in a $3f+1$ cluster.
- Ran pre-prepare, prepare, commit, and view-change phases while cutting per-request message complexity from quadratic toward linear, preserving safety and liveness on a seven-node cluster serving ten concurrent clients.

EDUCATION

Stony Brook University

August 2024 – May 2026

Master of Science in Computer Science and Applied Mathematics & Statistics

Stony Brook, New York

Guru Gobind Singh Indraprastha University

August 2017 – June 2021

Bachelor of Technology in Electronics and Communication

Paschim Vihar, Delhi

TECHNICAL SKILLS

Languages & CS: Python, Go, Java, C++, JavaScript, SQL, Data Structures, Algorithms, Object-Oriented Design, FastAPI
Systems & Distributed: Consensus (Paxos, 2PC, PBFT), Distributed & Parallel Systems, gRPC, Spark, Kafka, Unix/Linux
Cloud & DevOps: AWS (EMR, Lambda, S3, CloudWatch), CI/CD (GitHub Actions, Jenkins), Docker, REST, SDLC, Microsoft Azure, Kubernetes

Backend & Tools: API Design, Authentication & Authorization, Data Validation, Relational/NoSQL Databases, GenAI Dev Tools, Git, Monitoring, On-Call, Agile/Scrum, Automated Testing

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA-C03) | Amazon
Databricks Certified Data Engineer Associate | Databricks